UNIT QUESTIONS

1

SKILLS PROBLEM SOLVING (6)

a A fluorescent lamp is 25% efficient. Choose the statement that correctly describes how the lamp performs.

A the lamp only works for a quarter of the time

B 25% of the electrical energy transferred to the lamp is wasted

C the lamp does not work properly

D 25% of the energy supplied to the lamp is transferred to light **(1)**

SKILLS CRITICAL THINKING (5)

b A car uses energy stored in chemical form in petrol. When the engine is running this energy is transferred to other types of stored energy. Which of the following energy stores is useful?

A noise

B heat

C movement

D hot exhaust gases **(1)**

c Which of the following materials is a good thermal conductor?

A glass

B aluminium

C air

D water **(1)**

(6)

d Which of the following statements about thermal radiation is false?

A It cannot travel through glass.

B It can travel through a vacuum.

C It travels at the speed of light.

D It is absorbed by black objects better than shiny ones. **(1)**

SKILLS PROBLEM SOLVING (7)

e Four students, A, B, C and D, measure their power output by running upstairs and timing how long it takes. Which of the four students has the greatest power output?

A mass of 50 kg, takes 20 s to gain a height of 10 m

B mass of 40 kg, takes 30 s to gain a height of 15 m

C mass of 45 kg, takes 10 s to gain a height of 5 m

D mass of 55 kg, takes 25 s to gain a height of 15 m **(1)**

(Total for Question 1 = 5 marks)

SKILLS PROBLEM SOLVING

2 An electric motor is used to raise a load that weighs 800 N through a distance of 30 m.

7

a How much work does it do in raising the load? **(3)**

b Calculate the power output of the motor if it takes 16 s to raise the load. **(2)**

c The motor is 75% efficient.

SKILLS CRITICAL THINKING 6

i Explain what this means. **(2)**

SKILLS PROBLEM SOLVING 7

ii Calculate the electrical power that must be supplied to the motor to raise this load in the time stated. Give your answer in kW. **(2)**

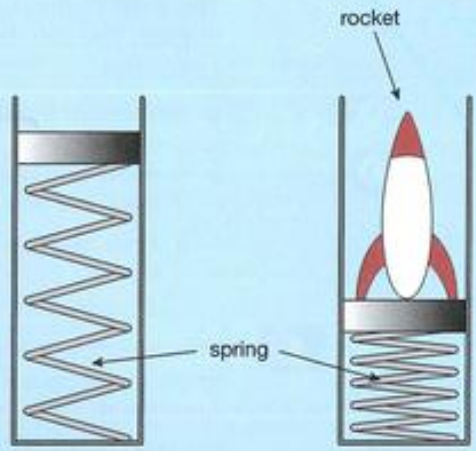
SKILLS INTERPRETATION

d Draw a labelled Sankey diagram to represent the energy transfers that take place as the motor raises the load. **(3)**

(Total for Question 2 = 12 marks)

SKILLS ANALYSIS, REASONING

3 The diagram shows a toy rocket launcher that uses a spring in a tube.



a Here are statements that describe the events leading to the launch of the toy rocket:

1 spring released

2 rocket leaves launcher

3 rocket gains kinetic energy

4 spring stores elastic potential energy

5 child does work squashing the spring

6 spring transfers stored energy to rocket

Choose from the following list the letter that describes the events in the order that they must happen. **(1)**

A 1 5 6 3 2 4 **B** 5 4 1 6 3 2 **C** 1 5 4 6 3 2 **D** 5 4 1 6 2 3

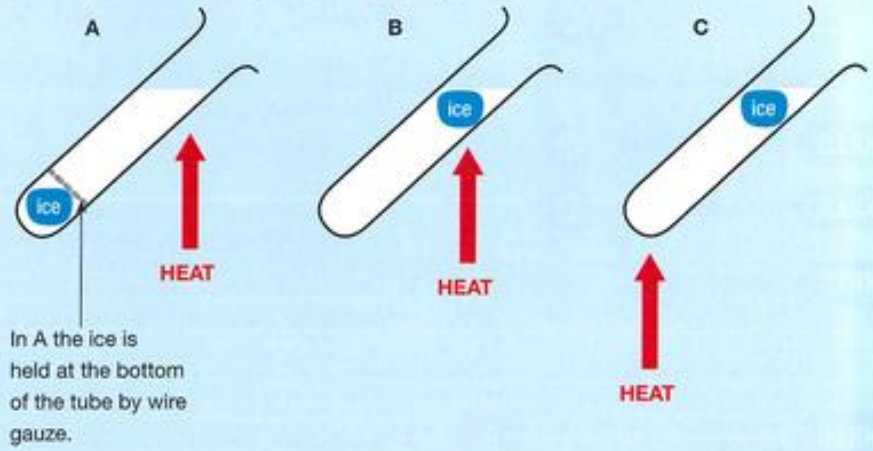
SKILLS CRITICAL THINKING

b Describe the energy transfers that take place after the rocket leaves the launcher, up to and including when it hits the ground. **(4)**

(Total for Question 3 = 5 marks)

SKILLS EXECUTIVE FUNCTION

4 A student wants to show that water is a poor conductor of heat. The diagram below shows the investigation he sets up.



The student measures how long it takes for each piece of ice in the water to melt. The results will help him to show whether or not water is a poor conductor.

4

a State anything the student should do to make sure this is a fair test. **(3)**

SKILLS REASONING

7

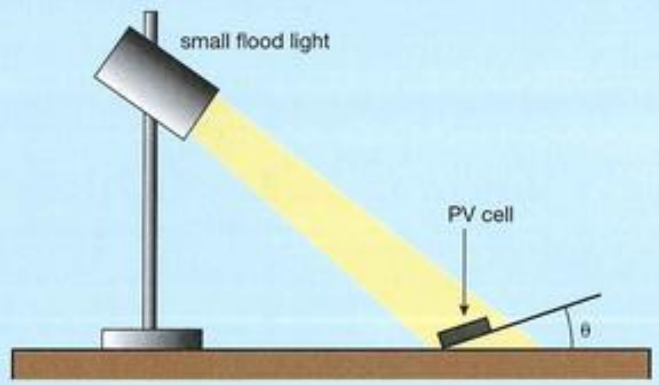
b i Assuming that this is a fair test, in which order do you think the ice will melt, starting with the slowest? Give an explanation for your answer. **(4)**

SKILLS EXECUTIVE FUNCTION

ii How can this investigation show that water is a poor conductor? **(3)**

(Total for Question 4 = 10 marks)

5 This question is about PV (photovoltaic) cells. A student wanted find the best angle to set up a PV cell to get the maximum amount of energy transferred from the Sun's rays to electrical energy. Here is the apparatus she set up and the results she obtained.



Angle θ/degrees	0	10	20	30	40	50	60	70	80	90
Current/mA	117	144	155	189	202	208	210	204	191	172

The PV cell was connected to a load and she measured the current, *I*, produced by the cell with the cell at different angles, θ, to the horizontal.

a State:

5

i the dependent variable **(1)**

ii the independent variable **(1)**

4

iii a control variable. **(1)**

SKILLS INTERPRETATION 7

b On a graph grid, 18 cm by 22 cm, plot a graph of the current produced by the PV cell against the angle of the cell to the horizontal. **(5)**

c Use the graph to find:

SKILLS PROBLEM SOLVING 5

i any anomalous result **(1)**

6

ii the angle to get the maximum output from the PV cell in the set-up shown. **(1)**

SKILLS EXECUTIVE FUNCTION

d When installing PV panels it is important to have them pointing towards the Sun as you would expect. In the experimental set-up shown, the lamp is modelling conditions in the UK in early March at midday (when the Sun reaches its highest position in the sky).

i How could the experiment be set up to model different positions of Sun? **(1)**

SKILLS INTERPRETATION 7

ii The experiment is repeated by another student at a time and place when the Sun is directly overhead at midday. Sketch a line on your graph to show how this student's results might appear. Label this line 'Sun overhead'. **(3)**

(Total for Question 5 = 14 marks)